JAN-P-106A

1 AUGUST 1946

SUPERSEDING
JAN-P-106
30 JUNE 1944

JOINT ARMY-NAVY SPECIFICATION

PACKAGING AND PACKING FOR OVERSEAS SHIPMENT—BOXES; WOOD, NAILED

(For weight of contents not in excess of 1,000 pounds)

Army Number
100-22A

Navy Number
39P16c
Section VI

This specification was approved by the War Department and the Navy Department for use of procurement services of the Army and the Navy and supersedes the following specifications:

U. S. Army	Navy Department
100–22	39P16b
30 June 1944	Section VI
	30 June 1944

A. APPLICABLE SPECIFICATIONS AND OTHER PUBLICATION

A–1. *Specifications.*—The following specifications, of the issue in effect on date of invitation for bids, form a part of this specification:

U. S. ARMY SPECIFICATIONS

94–40645—Marking; Shipment (Domestic and Export).[1]
100–2 —Standard Specification for Marking Shipments by Contractors.[2]

NAVY DEPARTMENT SPECIFICATION

General Specifications for Inspection of Material.[3]

FEDERAL SPECIFICATIONS

FF–N–101—Nails; Spikes; Staples; and Tacks.
FF–S–111—Screws; Wood.

A–2. *Other publication.*—The following publication, of the issue in effect on date of invitation for bids, forms a part of this specification:

BUREAU OF SUPPLIES AND ACCOUNTS PUBLICATION

Navy Shipment Marking Handbook.[3]

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[1] Applicable only to Army Air Forces purchases.
[2] Applicable only to Army purchases, except Army Air Forces.
[3] Applicable only to Navy purchases.

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B. STYLES

B–1. Nailed wood boxes covered by this specification shall be furnished in the following styles as specified in the contract or order. (See par. H–1.) When no style is specified, styles 2 to 4, inclusive, may be furnished. Style 5 boxes shall be furnished only when specified.

Style 1—Uncleated ends. (See fig. 1.)
Style 2—Full cleated ends, butt joints. (See fig. 2.)
Style 2½—Full cleated ends with horizontal cleats notches into vertical cleats. (See fig. 3.)
Style 3—Full cleated ends, mitered joints. (See fig. 4.)
Style 4—Two exterior vertical cleats on each end. (See fig. 5.)
Style 5—Two interior vertical cleats on each end. (See fig. 6.)

C. MATERIAL AND WORKMANSHIP

C–1. *Material.*

C–1a. *Wood.*—

C–1a(1). The species of wood which may be used for lumber for nailed wood boxes are classified in groups as follows, and when a group is specified, any species in that group may be used at the option of the contractor:

Group 1

Aspen (popple)
Basswood
Buckeye
Butternut
Cedar
Chestnut
Cottonwood
Cucumbertree
Cypress
Fir, alpine
Fir, balsam
Fir, noble
Fir, white
Magnolia
Pine, jack
Pine, lodgepole
Pine, ponderosa (western yellow)
Pine, red (Norway)
Pine, sugar
Pine, white
Redwood
Spruce
Willow
Yellow poplar

Group II

Douglas fir
Hemlock
Larch (tamarack)
Pine, southern

Group III

Ash, black
Ash, pumpkin
Blackgum
Elm, white
Maple, soft
Sweetgum (red gum)
Sycamore
Tupelo, water

Group IV

Ash, white
Beech
Birch
Elm, rock
Hackberry
Hickory
Maple, hard
Oak
Pecan

C–1a(2). Lumber shall be seasoned to a moisture content not to exceed 18 percent, not less than 12 percent of its oven-dry weight. At least one surface, i.e., the surface to be placed on the outside of

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the box, shall be sufficiently smooth to permit legible marking, stenciling or printing. Pieces shall be cut true to length. The boards shall be free from all defects that materially weaken them, expose the contents of the box to damage, or interfere with the prescribed fabrication or nailing. No knot, knot cluster, or series of knots across the width of the board, within a length equal to the width of the board shall have a diameter or a sum of diameters greater than one-third the width of the board.

C–1a(3). The divergence of the grain in each cleat shall not exceed 1 inch in 10 inches of length.

C–1b. *Nails.*—Nails shall comply with the requirements of Federal Specification FF–N–101. Nails shall be coolers, corkers, sinkers, standard box, or commons. (See par. H–3.)

C–1b(1). *Unclinched nails.*—All unclinched nails shall be either cement-coated or chemically etched. Chemically etched nails shall be of standard box or common type, or other suitable types, if approved by contracting agency, and shall have the surface roughened by chemical action. (See par. H–4 for suggested procedure for etching nail surfaces.)

C–1b(2). *Clinched nails.*—Clinched nails shall be cement-coated, chemically etched, or bright uncoated. Nails shall be of sufficient length to pass through cleat (or batten) and end (or side, top, or bottom) and be clinched not less than ⅛ inch.

C–1c. *Corrugated fasteners.*—The sizes of corrugated fasteners used shall conform to the requirements of table I.

TABLE I.—*Size of corrugated fasteners for boards.*

Thickness of box part	Size of corrugated fasteners
Inches	*Inch*
7/16 to 31/64, incl.	1/4 x 1
3/8 to 39/64, incl.	5/16 x 1
1/2 to 39/64, incl.	3/8 x 1
5/8 to 51/64, incl.	1/2 x 1
7/8 and up.	5/8 x 1
	3/4 x 1

C–1d. *Screws.*—Screws shall comply with the requirements of Federal Specification FF–S–111, standard flat head.

C–2. *Workmanship.*—Boxes shall be well manufactured in accordance with high grade commercial practice. The boxes shall be free from imperfections which may affect the utility.

D. GENERAL REQUIREMENTS

D–1. *Container manufacturer's identification.*—When specified in the contract or order (see par. H–1), each box shall be marked with the specification number, the name and address of the container supplier, and the maximum weight of contents and type of load that may be packed in the container. All these markings shall be limited to 15 square inches in area and shall be placed in a lower corner of one side panel in letters approximately 5/16-inch

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high, except that "JAN–P–106" shall be in letters approximately ¾-inch high. Arrangement of these markings shall be as follows:

COMPLIES WITH SPEC.
JAN–P–106
JOHN DOE BOX CO.
CHICAGO, ILLINOIS
FOR MAXIMUM WT. CONTENTS LBS.
FOR TYPE LOAD

When boxes are supplied in shook form by the container manufacturer, the first line of the identification may be changed to read: SHOOK COMPLIES WITH SPEC.

D–2. *Thickness and width of boards.*—

D–2a. *Thickness.*—The minimum thickness of the ends, sides, top, and bottom and the minimum thickness and width of cleats, shall be determined from tables II, III, and IV. Occasional variations in thickness due to mismanufacture will be permitted in not more than 10 percent of the pieces, but no part of any piece shall be less than ⅞ of the required thickness. If boards ⅜ inch or more in thickness are surfaced on both sides (to protect the contents), the thickness may be 1/32 inch less than the thickness as determined from tables II, III, and IV.

D–2b. *Width.*—The number of pieces in any side, top, bottom, or end shall be chosen so that no piece shall be less than 2½ inches in width. One piece is considered to be either a single unjointed part or two or more pieces jointed or matched in accordance with paragraphs D–2b(1) or D–2b(2). Each cleat or batten shall be a single unjointed piece of wood.

D–2b(1). Two or more pieces that are Linderman-jointed and glued shall be considered one piece. Linderman joints consist of a projection along the edge on one piece extending into a corresponding recess in the edge of the adjoining part, the projection and the recess extending the length of the pieces. Linderman joints differ from tongue and grooved joints in that a cross section of the joint is a dove tail; i.e., the outer edge of the projection is thicker than the inner edge. Linderman-jointed pieces are assembled by sliding the projection longitudinally into the recess from the end. The glue used in Linderman-jointed pieces shall be water-resistant and toxic treated.

D–2b(2). Two or more pieces ⅜ inch or more in thickness and not less than 1½ inches in width at either end, which are butt-jointed and matched side to side the butt joint being fastened with corrugated fasteners, shall be considered one piece when designed for the sides, top or bottom of the box or for ends of style 1 boxes. Butt joints will not be permitted on the ends of styles 2, 2½, 3, 4, and 5 boxes. Fasteners shall be spaced not more than 8 inches apart. Sizes of fasteners required are shown in table I. If three or more corrugated fasteners are required in a joint, they shall be driven alternately from opposite sides of the part, unless the joint tongued-and-grooved and glued in which case fasteners may be ,ven from one side.

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TABLE II.—*Determination of sizes of nails and minimum thickness of ends, sides, tops and bottoms for style 1 boxes.* (See appendix I, par. 2–a.)

Minimum thickness of sides, top and bottom, all groups of woods	Minimum thickness of ends, all groups of woods[1]	Size of nails for nailing sides, tops and bottoms to ends			
		Groups of woods			
		I	II	III	IV
Inch	*Inch*				
11/32	3/4	7d	6d	6d	5d

[1] The end shall be made from one piece.

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TABLE III.—*Type 1 (easy) and type 2 (average) load—determination of size of nails, minimum thickness of sides, tops, bottoms and ends, and minimum thickness and width of cleats for styles 2, 2½, 3, 4, and 5 nailed wood boxes.*

Weight of contents	Style of box[1]	Groups I and II woods					Groups III and IV woods				
		Minimum thickness of sides, tops and bottoms	Size of nail for nailing sides, tops and bottoms to end and cleats[2]		Minimum thickness of ends	Minimum thickness and width of cleats	Minimum thickness of sides, tops and bottoms	Size of nails for nailing sides, tops and bottoms to ends and cleats[2]		Minimum thickness of ends	Minimum thickness and width of cleats
			Group I	Group II				Group III	Group IV		
Pounds		*Inch*			*Inch*	*Inches*	*Inch*			*Inch*	*Inches*
To 50, incl.	4 or 5	3/8	6d	5d	5/8	5/8×1-3/4	5/16	5d	5d	5/8	5/8×1-3/4
51 to 100, incl.	4 or 5	7/16	7d	6d	3/4	3/4—2-1/4	3/8	5d	5d	5/8	5/8×1-3/4
101 to 250, incl.	4 or 5	9/16	8d	7d	3/4	3/4—2-1/4	1/2	6d	5d	11/16	11/16×2-1/4
101 to 250, incl.	2, 2½ or 3	9/16	8d	7d	5/8	5/8—2-1/4	1/2	6d	5d	5/8	5/8×2-1/4
251 to 400, incl.	4 or 5	11/16	9d	8d	25/32	25/32×2-5/8	9/16	6d	6d	3/4	3/4×2-1/4
251 to 400, incl.	2, 2½ or 3	11/16	9d	8d	3/4	3/4—2-5/8	9/16	6d	6d	11/16	11/16×2-1/4
401 to 600, incl.	2, 2½ or 3	25/32	9d	8d	25/32	25/32×2-5/8	5/8	6d	6d	3/4	3/4×2-1/4
601 to 1,000, incl.	(See table III.)										

[1] When the inside depth of a box is 5 inches or less, end cleats shall not be used. However each side and each end shall be made from one piece, except when the end is approximately square, a two-piece end with grain running at right angles to each other may be used. Both pieces used for double ends shall be approximately equal thickness. Thickness of the ends shall be not less than the combined thickness of the end and cleat as specified for style 4 boxes.

[2] When minimum thickness shown is exceeded, nail sizes shall be increase to correspond.

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TABLE IV.—*Type 3 (difficult) load—determination of size of nails, minimum thickness of sides, tops, bottoms and ends and minimum thickness and width of cleats for styles 2, 2½, 3, 4, and 5 nailed wood boxes.*

Weight of contents	Style of box[1]	Groups I and II woods					Groups III and IV woods				
		Minimum thickness of sides, tops and bottoms	Size of nails for nailing sides, tops, and bottoms to ends and cleats[2]		Minimum thickness of ends	Minimum thickness and width of cleats	Minimum thickness of sides, tops and bottoms	Size of nails for nailing sides, tops, and bottoms to ends and cleats[2]		Minimum thickness of ends	Minimum thickness and width of cleats
			Group I	Group II				Group III	Group IV		
Pounds		*Inches*			*Inch*	*Inches*	*Inch*			*Inches*	*Inches*
To 100, incl.	4 or 5	1/2	8d	7d	3/4	3/4×2-1/4	7/16	5d	5d	5/8	5/8×1-3/4
To 100, incl.	2, 2½ or 3	1/2	7d	6d	5/8	5/8×2-1/4	7/16	5d	5d	5/8	5/8×1-3/4
101 to 250, incl.	4 or 5	5/8	8d	7d	25/32	25/32×2-5/8	1/2	6d	5d	3/4	3/4×2-1/4
101 to 250, incl.	2, 2½ or 3	5/8	8d	7d	3/4	3/4×2-1/4	1/2	5d	5d	5/8	5/8×1-3/4
251 to 400, incl.	4 or 5	3/4	10d	9d	1-1/16	1-1/16×3-1/4	5/8	7d	6d	13/16	13/16×2-3/4
251 to 400, incl.	2, 2½ or 3	3/4	9d	8d	3/4	1-1/16×3-1/4	5/8	6d	5d	3/4	3/4×2-3/4
401 to 600, incl.	2, 2½ or 3	25/32	9d	8d	25/32	1-1/16×3-1/4	11/16	7d	6d	13/16	13/16×2-3/4
601 to 800, incl.	2, 2½ or 3	13/16	10d	9d	1-1/16	1-1/16×3-1/4	3/4	8d	7d	13/16	13/16×2-3/4
801 to 1,000, incl.	2, 2½ or 3	1-1/16	12d	12d	1-5/16	1-5/16×4-1/8	7/8	9d	8d	1-1/16	1-1/16×3-3/8

[1] When the inside depth of a box is 5 inches or less, end cleats shall not be used. However each side and each end shall be made from one piece, except when the end is approximately square, a two-piece end with grain running at right angles to each other may be used. Both pieces used for double ends shall be approximately equal thickness. Thickness of the ends shall be not less than the combined thickness of the end and cleat as specified for style 4 boxes.

[2] When minimum thickness shown is exceeded, nail sizes shall be increased to correspond.

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D–3. *Additional cleats or battens.*—Additional cleats or battens, of the same width and thickness as on the ends of the box, shall be applied to sides, top and bottom when the unsupported span exceeds that given in table V for the thickness of the part involved. Additional cleats applied to ends shall run across the grain of the end boards and be placed intermediate between the regular cleats. Battens applied to sides, top and bottoms shall run across the grain of the part to which they are applied and shall be placed inside the box whenever the nature of the contents will permit. When battens are required and are placed on the outside of the box, not less than two sets shall be attached to the outer faces.

TABLE V.—*Requirements for additional cleats or battens.*

Thickness of end side, top or bottom		Length of unsupported span
Groups I and II woods	Groups III and IV woods	
Inches	*Inch*	*Inches*
3/8 and 7/16	5/16 and 3/8	23
1/2	7/16	30
5/8	1/2	38
3/4	5/8	45
25/32	11/16	47
13/16	3/4	50
1-1/16	7/8	64

D–4. *Nailing.*—

D–4a. The size of nails for fastening sides, top and bottom to ends and cleats, as determined by the species of wood and the thickness of the parts fastened together, shall conform to the requirements shown in tables II, III and IV.

D–4b. The size of nails driven through the top and bottom into sides of the box shall conform to the requirements shown in table VI.

TABLE VI.—*Sizes of nails for fastening top and bottom to sides.*

Thickness of side	Group I wood	Group II wood	Groups III and IV woods
Inches			
Under 3/4		No nailing permitted	
3/4 to 7/8. incl.	7d	6d	5d
15/16 to 1-1/16, incl.	8d	7d	....

D–4c. The average spacing of nails holding the sides, top or bottom to the ends and cleats or battens shall be not greater than the values shown in table VII. The first and last nails shall be spaced one-half of the specified spacing from the ends of the nailing edge, but not less than ¾ inch. In the event that it is necessary to exceed this spacing because of small knots or checks in the nailing and or because of the location of joints between boards, the dis-

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tance between any two adjacent nails shall not be greater than one and one-half times the spacing given in table VII.

D–4d. If the desired nail is not available, one size smaller shall be used and nails spaced ¼ inch closer than is required for the size of nail substitutes.

D–4e. Each board in the sides, top, or bottom shall have at least two nails at each nailing end.

D–4f. Wherever cleats are used in the end construction, approximately one-half of the nails used to secure lengthwise boards, forming top, bottom, or sides to cleat edge shall be driven into end and the remainder into cleat. For nailing sides to ends and cleats, the nail spacing shall be as shown in table VII.

D–4g. When the top and bottom are nailed to the sides, the nails shall be spaced between 6 and 8 inches apart.

D–4h. Each piece of the end shall be nailed to each vertical cleat with not less than two nails.

D–4i. The nails shall pass through both the cleat (or batten) and the end (or side, top, or bottom) and be clinched not less than ⅛ inch. The nails in each cleat shall be driven in two rows spaced as given in table VII and located as shown on figures 1, 2, and 3. Nailing of cleats or battens to sides, top, and bottom shall be in accordance with the same requirements.

TABLE VII.—*Average spacing of nails (except for nailing top and bottom to sides).*

Size of nails	Average spacing when driven into cleat and side grain of end	Average spacing when driven into cleat and end grain of end
	Inches	*Inches*
6d or smaller	2	1-3/4
7d	2-1/4	2
8d	2-1/2	2-1/4
9d	2-3/4	2-1/2
10d	3	2-3/4
12d	3-1/2	3
16d	4	3-1/2
20d	4-1/2	4

D–5. *Use of screws.*—

D–5a. *General.*—Screws shall be used for fastening boxes only when so specified.

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D–5b. Screw length shall be three times the thickness of the piece holding the head. If this length is not available, use the next shorter standard length. The gage of screws to be used shall be determined from table VIII. If the required size of screw is unavailable, use next smaller gage and reduce spacing ¼ inch.

TABLE VIII.—*Gage of screws for different thicknesses of lumber and groups of wood species.*

Thickness of piece holding point of screw	Screw gage for wood groups			
	I	II	III	IV
Inches	*Number*	*Number*	*Number*	*umber*
3/8	6	5	5	4
1/2	7	6	6	5
5/8	8	6	6	6
11/16	8	7	6	6
3/4	9	8	7	7
25/32	9	8	8	7
13/16	10	9	8	8
7/8	10	9	8	8
1	11	10	9	9
1-1/16	12	10	10	9
1-1/8	12	11	10	10
1-1/4	12	12	11	11
1-5/16	12	12	12	12

D–5c. *Spacing of screws.*—The spacing of screws for fastening sides, top, and bottom to ends and cleats shall be determined from table IX. When top and bottom are screwed to the sides, screws shall be placed 8 to 12 inches apart.

TABLE IX.—*Average spacing of screws for fastenings to ends and cleats.*

Gage of Screws	Spacing for assembling sides, top and bottom		Spacing when only top of box is fastened with screws
	When driven into cleats and side grain of ends	When driven into cleats and end grain of ends	
Number	*Inches*	*Inches*	*Inches*
7 or smaller	2	1-3/4	3
8	2-1/4	2	3-1/4
9	2-1/2	2-1/4	3-1/2
10	2-3/4	2-1/2	3-3/4
11	3	2-3/4	4
12	3-1/2	3	4-1/2

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D–5d. *Driving of screws.*—Shank clearance holes shall be drilled in the pieces holding the shanks of screws. Pilot holes, drilled in the piece holding the points of screws, are required for group IV wood. Table X gives a guide for sizes of twist bits or drills to be used in drilling shank clearance holes and pilot holes for screws. Screws shall be hammer-driven in groups I, II, and III woods only far enough to make threads take hold, then driven the remaining distance with a screwdriver, avoiding overdriving.

TABLE X.—*Sizes of bits for drilling shank clearance holes and pilot holes for screws.*

Gage of screw	Shank clearance holes [1]		Pilot holes for group IV wood [2]	
	Size of twist bit	Gage of drill	Size of twist bit	Gage of drill
Number	*Inch*	*Number*	*Inch*	*Number*
4	7/64	34	1/16	52
5	1/8	31	5/64	49
6	9/64	29	5/64	47
7	9/64	25	3/32	44
8	5/32	20	3/32	40
9	11/64	16	7/64	37
10	3/16	12	7/64	33
11	13/64	7	1/8	31
12	7/32	3	1/8	30

[1] If size of twist bit or drill shown is not available, use next smaller size for groups I and II woods, next larger size for groups III and IV woods.

[2] Use next smaller size twist bit or drill shown for groups I, II, or III woods.

D–5e. *Countersinking.*—All screws shall be so driven that their heads shall not protrude above the surface of the board. Countersinking for screw heads is not necessary in groups I and II woods, but is desirable in groups III and IV woods.

D–6. *Fabrication.*—

D–6a. All parts of the box shall be cut square to size, and the box shall be fabricated in accordance with good commercial practice.

D–6b. Nails shall be driven so that neither the head nor the point will project above the surface of the wood. Occasional overdriving of nails will be permitted but no nail shall be overdriven more than ⅛ the thickness of the piece.

D–7. *Closure and strapping.*—When boxes covered by this specification are furnished to the Government as filled containers, the closure and strapping of boxes shall be as specified in appendix I.

E. DETAIL REQUIREMENTS

E–1. *Cleats.*—

E–1a. *Styles 2 and 2½.*—The ends of the cleats which are positioned across the grain of the end board shall be from ⅛ inch to 1/16 inch from the inside surface of the top and bottom. The sides, top, and bottom shall extend over the cleats.

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E–1b. *Style 3.*—The ends of the cleats shall be mitered at 45°. Cleats shall be flush with the edges of the end panel. The sides, top and bottom shall extend over the cleats.

E–1c. *Style 4.*—The cleats shall be positioned across the grain of the end boards and shall extend within ⅛ inch of the outside surface of the top and bottom. Only the side shall extend over the cleats. Alternatively, this style of box may be constructed with two horizontal cleats at each end, one at the top edge and the other at the bottom edge of each end. In all cases, the grain of the ends shall be at right angles to the grain of the cleats. The top and bottom shall extend over the sides. Handle cleats, when used on style 4 box, shall be of the same construction as the other cleats and shall be placed one on each end of the box, horizontally. Bottom edges of cleats shall be located not more than ⅓ of the box depth from the top edge of the end. Handle cleats shall make butt-joints with the vertical cleats.

E–1d. *Style 5.*—The cleats may have either a triangular or rectangular cross-section. Triangular cleats shall have not less than the same cross-sectional area required for corresponding rectangular cleats. The cleats shall be positioned across the grain of the end boards and shall extend within ⅛ inch of the inside surface of the top and bottom. The sides shall extend over the ends and be flush with the outside face of the ends.

F. METHODS OF SAMPLING, INSPECTION, AND TESTS

F–1. *Sampling.*—Samples of any materials, components, etc., not furnished by the Government, entering into the manufacture of the articles covered herein shall be selected from time to time by the Government inspector, carefully examined and tests made to determine if they are in accordance with the requirements of this specification or as specified by the contracting officer.

F–2. *Inspection.*—Inspection may be made throughout the entire process of manufacture. The passing as satisfactory of any detail of construction or materials shall not relieve the contractor of responsibility for faulty workmanship or materials which may be discovered at any time prior to final acceptance. For Army purchases, final inspection of the finished article shall be made either at place of manufacture or at point of delivery designated in the contract or order. (See par. H–1.) In case of factory inspection, every facility shall be afforded inspectors, by the manufacturers, for the prosecution of their work. For Navy purchases, final inspection shall be made at the place of manufacture, unless otherwise specified.

G. PACKAGING, PACKING, AND MARKING FOR SHIPMENT

G–1. Unless otherwise specified, boxes shall be shipped either completely assembled except for the tops, which shall be bundled or crated as complete panels; or knocked down (except ends shall be cleated) and panels of like size securely bundled or crated in quantities that permit easy loading and handling.

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G–2. *Marking.*—In addition to any special marking required by the contract or order, shipments for the Army, except Army Air Forces, shall be marked in accordance with the requirements of U. S. Army Specification 100–2; for Army Air Forces in accordance with the requirements of U. S. Army Specification 94–40645; for the Navy in accordance with the requirements of the Navy Shipment Marking Handbook.

H. NOTES

H–1. Requests, requisitions, schedules, and contracts or orders should specify the following:

(a) Number and date of the specification.
(b) Quantity required.
(c) Weight of contents.
(d) Style of box required. (See par. B–1.)
(e) Inside dimensions of box specified in inches to the nearest sixteenth of an inch in order of length by width by depth.
(f) Container manufacturer's identification, if required. (See par. D–1.)
(g) Place of inspection. (See par. F–2.)
(h) Whether to be shipped assembled or knocked down. (See par. G–1.)
(i) Type of load. (See par. H–5.)

H–2. Exceptional commodities or especially dangerous articles may require better boxes than are covered by this specification. In every case, the containers must conform to the specifications prescribed in Interstate Commerce Commission regulations governing the transportation of the particular articles being packed.

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H-3. *Nails.*—The sizes and weights of nails complying with the requirements of Federal Specification FF-N-101 and as specified in paragraph C-1b are given in tables XI and XII.

TABLE XI.—*Size and weight of cement-coated box nails.*

Size	Length	Coolers			Corkers		
		Diameter	Head	Number per pound	Diameter	Head	Number per pound
	Inches	*Inch*	*Inch*		*Inch*	*Inch*	
4d	1-3/8	.0800	7/32	488	.0860	7/32	392
5d	1-5/8	.0860	15/64	364	....	.....	....
6d	1-7/8	.0915	1/4	275	.0990	1/4	232
7d	2-1/8	.0990	17/64	212	....	.....	....
8d	2-3/8	.1130	9/32	144	.1205	9/32	129
9d	2-5/8	.1130	9/32	127	.1205	9/32	114
10d	2-7/8	.1205	19/64	104	.1350	5/16	84
12d	3-1/8	....	.....	...	.1350	5/16	77
16d	3-3/8	....	.....	...	.1483	11/32	59
20d	3-7/8	....	.....	...	.1770	3/8	36

Size	Length	Sinkers			Standard		
		Diameter	Head	Number per pound	Diameter	Head	Number per pound
	Inches	*Inch*	*Inch*		*Inch*	*Inch*	
4d	1-3/8	.0800	13/64	488	.0670	13/64	710
5d	1-5/8	.0860	7/32	364	.0720	7/32	536
6d	1-7/8	.0915	15/64	275	.0860	1/4	306
7d	2-1/8	.0990	1/4	212	.0860	1/4	268
8d	2-3/8	.1130	17/64	142	.0990	17/64	186
9d	2-5/8	.1130	17/64	130	.0990	17/64	167
10d	2-7/8	.1205	9/32	104	.1130	19/64	118
12d	3-1/8	.1350	5/16	77	....	.....	....
16d	3-1/4	.1483	11/32	61	....	.....	....
20d	3-3/4	.1770	3/8	37	....	.....	....

TABLE XII.—*Size and weight of bright (uncoated) standard box and common nails.*

Size	Length	Standard			Common	
		Diameter	Head	Number per pound	Diameter	Number per pound
	Inches	*Inch*	*Inch*		*Inch*	
4d	1-1/2	.0800	7/32	473	.0985	316
5d	1-3/4	.0800	7/32	406	.0985	271
6d	2	.0990	17/64	236	.113	181
7d	2-1/4	.0990	17/64	210	.113	161
8d	2-1/2	.1130	19/64	145	.131	106
9d	2-3/4	.1130	19/64	132	.131	96
10d	3	.1280	5/16	94	.1483	69
12d	3-1/4	.1280	5/16	88	.1483	63

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H–4. *Chemically etched nails.*—The following procedure is suggested for chemically etching nails:

a. Prepare a 10-percent solution (by weight) of commercial monoammonium phosphate in water. Do *not* use metal container for preparing or storing solution. Keep solution near room temperature (about 68°F).

b. Immerse nails in solution for about 7 hours, stirring occasionally. Five gallons of solution is sufficient to etch about 100 pounds of nails.

c. At end of etching period, remove nails and rinse with water.

d. Air dry nails to prevent rusting.

H–5. *Type of load.*—The design of shipping box to be used is determined by the type of load. The type of load is determined by the contents' weight and size, fragility, shape, and capacity for support of or damage to the box. The classification of type of load shall be as follows:

H–5a. *Type 1 (easy) load.*—

1. Contents are not easily damaged by shock or puncture, and are durable enough to withstand the forces encountered in transportation and handling.
2. Contents are of low or moderate density.
3. Contents are prepacked in one container which completely fills supports, and adds rigidity to all faces of the shipping box, or
4. Contents completely fill, support, and add rigidity to the shipping box, which is used to provide surface protection and facilitate handling and/or storage. Contents cannot shift or flow.

Examples: Solid material (one piece): boxed articles, paper, or textiles; chest or kit of tools, wooden cabinet, boxed sturdy instruments.

H–5b. *Type 2 (average) load.*—

1. Contents are packed directly into outer shipping box, or
2. Contents are protected by interior packing such as wrapping, partitions, pads, or fiberboard boxes, and
3. Contents or intermediate packing provide support at several points on each face of the shipping box.
4. Contents are of low or moderate density.

Examples: Goods in metal cans which are not packed in an inner container, bottles individually cushioned, hardware and other articles in cartons.

H–5c. *Type 3 (difficult) load.*—

1. Contents require high degree of protection from puncture, shock, or distortion of container, or
2. Contents do not support all faces of the shipping box, or
3. Contents are of high density.

Examples: Long bolts and rods which can exert highly concentrated force on two faces of the shipping box or roll against other faces. Dense articles such as rivets and drop forgings which are free to shift or flow and concentrate force on one box face. Articles requiring special

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protection because they are delicate, dangerous, or easily punctured (instruments, explosives, acids, etc.).
Machined parts and assemblies that do not completely fill the shipping box and require bracing, bolting, cushioning, or "floating".

H–6. Copies of Joint Army-Navy specifications and Federal specifications (required for Army purchases) and U. S. Army specifications may be obtained, as indicated in the "Index of United States Army, Joint Army-Navy, and Federal Specifications Used by the War Department". Copies of this Index may be obtained from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Agencies within the War Department will obtain copies of Joint Army-Navy, United States Army, and Federal specifications through established War Department channels. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

H–7. Copies of Joint Army-Navy specifications and Federal specifications (required for Navy purchases), Navy Department specifications, and the Navy Shipment Marking Handbook may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that Naval activities should make application to the Supply Officer in Command, Naval Supply Depot, Bayonne, N. J. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

H–8. Copies of this Joint Army-Navy specification (required for Army purchases) may be obtained from the following points:

Chicago Quartermaster Depot, 1819 West Pershing Road, Chicago 9, Ill.
Stockton General Depot, U. S. Army, Stockton 1, Calif.
San Antonio General Depot, U. S. Army, Grayson Street Station, San Antonio, Tex.
Jeffersonville Quartermaster Depot, Jeffersonville, Ind.
Philadelphia Quartermaster Depot, 2800 South Twentieth Street, Philadelphia 45, Pa.
Kansas City Quartermaster Depot, 601 Hardesty Avenue, Kansas City 1, Mo.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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APPENDIX I

USE OF NAILED WOOD BOXES AND REQUIREMENTS FOR CLOSURE AND STRAPPING

1. APPLICABLE SPECIFICATIONS

1–a. The following specifications, of the issue in effect on date of invitation for bids, form a part of this appendix.

NAVY DEPARTMENT SPECIFICATION

General Specifications for Inspection of Material.[1]

FEDERAL SPECIFICATIONS

QQ–S–781—Strapping, Flat; Steel
QQ–S–790—Strapping, Round; Steel, Bare and Zinc-coated.

2. USE

2–a. *Style 1 boxes.*—Style 1 boxes shall be used only for canned goods and subsistence items not packed in glass, and only when specified by the procuring agency. The weight of contents for these boxes shall not exceed 60 pounds; the inside length, width and depth shall not exceed 20 inches, 14 inches, and 10½ inches respectively.

2–b. *Styles 2, 2½, and 3 boxes.*—Styles 2, 2½ and 3 boxes shall not be used for any weight of contents exceeding 1,000 pounds, except with approval of procuring agency.

2–c. *Styles 4 and 5 boxes.*—Styles 4 and 5 boxes shall not be used for weight of contents exceeding 400 pounds. Style 5 boxes shall be used only when specified.

3. CLOSURE

3–a. *Nails.*—For requirements of nails, see paragraph C–1b. Size and spacing of nails shall be in accordance with paragraph D–4.

3–b. *Screws.*—For requirements covering screws, see paragraph C–1d. Size and spacing of screws shall be in accordance with paragraph D–5.

4. STRAPPING

4–a. *General.*—Strapping shall be performed by the activity by which the box is packed. The strapping is not to be furnished by the box maker unless specifically required by the contract or order. All strapping shall be nailless (type I), treated to resist rust, flat steel or class A or B galvanized round wire, complying with the requirements of Federal Specifications QQ–S–790 and QQ–S–781, unless otherwise specified.

[1] Applicable only to Navy purchases.

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4–b. *Size.*—Size of round wire strapping used shall be determined from table I–a and the size of flat strapping used shall be determined from table II–a.

TABLE I–a.—*Minimum gage of round wire for various weights of boxes.*

Net weight of contents		Gage of wire when different number of wires are used			
		Two bands		Three bands	
		100,000 p.s.i. tensile strength	140,000 p.s.i. tensile strength	100,000 p.s.i. tensile strength	140,000 p.s.i. tensile strength
Pounds		*Inch*	*Inch*	*Inch*	*Inch*
To 70		0.0720 (15 gage)	.0625 (16 gage)	.0720 (15 gage)	.0625 (16 gage)
70 to 125,	incl. all styles	.0800 (14 gage)	.0720 (15 gage)	.0800 (14 gage)	.0720 (15 gage)
126 to 175,	incl. " "	.0915 (13 gage)	.0800 (14 gage)	.0915 (13 gage)	.0800 (14 gage)
176 to 250,	incl. " "	.0915 (13 gage)	.0915 (13 gage)	.0915 (13 gage)	.0915 (13 gage)
251 to 400,	incl. " "	.1055 (12 gage)	.0990 (12½ gage)	.0915 (13 gage)	.0915 (13 gage)
401 to 1,000,	incl. " "	.....	.....	.1055 (12 gage)	.0990 (12½ gage)

TABLE II–a.—*Minimum sizes of flat metal bands for various weights of boxes.*

Net weight of contents		Dimensions of flat metal bands when different numbers of bands are used.	
		Two bands	Three or more bands
Pounds		*Inch*	*Inch*
To 70		½ x 0.015	½ x .015
70 to 125,	incl. all styles	½ x .020	½ x .020
126 to 175,	incl. " "	½ x .020	½ x .020
176 to 250,	incl. " "	½ x .020	½ x .020
251 to 400,	incl. " "	½ x .020	½ x .020
401 to 1,000,	incl. " "	.....	½ x .023

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4–c. *Tensile strength.*—The ultimate tensile strength of flat strapping shall be not less than 80,000 p.s.i. The strength of the joint shall be at least 75 percent of the tensile strength of the flat or round wire strapping. The strength of joints shall be computed from the average required tensile strength of three specimens of jointed straps and three specimens of strapping. These specimens may be any convenient length. Each joint specimen shall be prepared by fastening together the ends of two pieces of strapping with the joint made in the usual way. The joint shall be near the middle of the specimen. A standard tensile strength testing machine approved by the inspector shall be used to determine the strength of each specimen.

4–d. *Tightness.*—All straps shall be applied perpendicular to the edges of the box over which they pass and shall be drawn tight so as to sink into the wood at the edges. Straps shall be applied just prior to shipment where practicable.

4–e. *Distance of straps from box ends.*—If two or more straps are used, the distance between end straps and the end of the box shall be approximately one-sixth the length of the box. The intermediate straps shall be spaced equally between the end straps.

4–f. *Number of straps to be used.*—The number of straps to be used shall agree with the number indicated in tables I–a and II–a for the weight and contents listed. For styles 2, 2½, 3, 4 and 5 boxes, two or more girth-wise straps shall be applied except that only one girth-wise strap shall be applied when weight is less than 35 pounds and length does not exceed 12 inches. When the outside length of the box exceeds 36 inches, three or more girth-wise straps shall be applied so that the distance between the straps shall be not more than 24 inches. Style 1 box and box permitted under note 1 to tables III and IV shall have one strap applied around the top, bottom, and ends of the box and after this strap has been applied, two additional straps shall be applied girth-wise (around the top, bottom, and sides of the box).

4–g. *Staples.*—

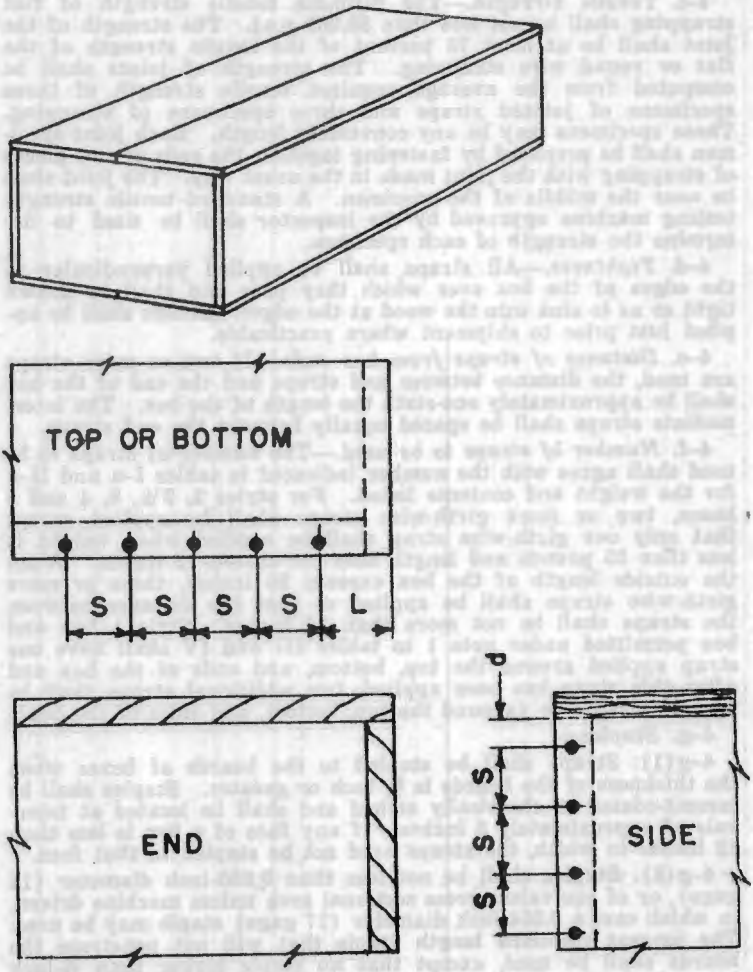
4–g(1). Straps shall be stapled to the boards of boxes when the thickness of the boards is ⅝ inch or greater. Staples shall be cement-coated or chemically etched and shall be located at intervals of approximately 6 inches. If any face of a box is less than 12 inches in width, the straps need not be stapled to that face.

4–g(2). Staples shall be not less than 0.080-inch diameter (14 gage), or of equivalent cross sectional area unless machine driven, in which case a 0.054-inch diameter (17 gage) staple may be used. The longest standard length staple that will not penetrate the boards shall be used, except that no staple longer than ¾-inch is required. Staples shall not be overdriven so as to cause damage to strapping.

5. INSPECTION

5–a. Inspection of boxes, for commodities required to be packed for shipment in boxes in accordance with this specification, may consist of visual examination together with measurement of the component parts of the box. Tests for tensile strength of bands and wire shall be at the option of the Government inspector.

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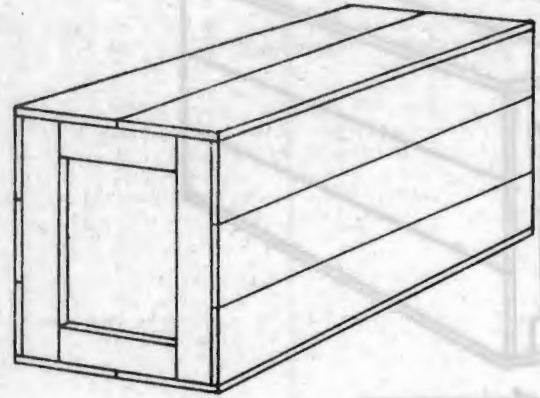


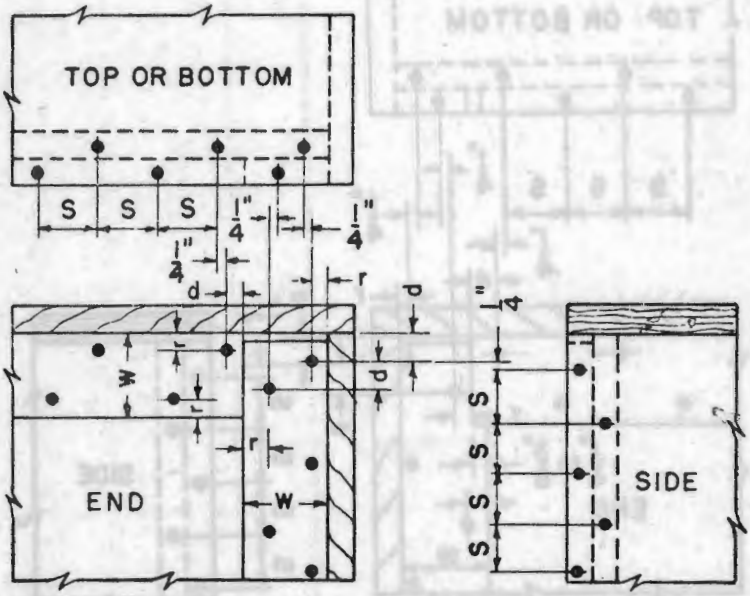
NOTES:
S = Nail spacing. (See par. D–4.)
L = Length of nails for nailing sides to ends.
d = ¾ inch.

FIGURE 1.—*Style 1 box (uncleated ends).*

[JAN–J–106A]

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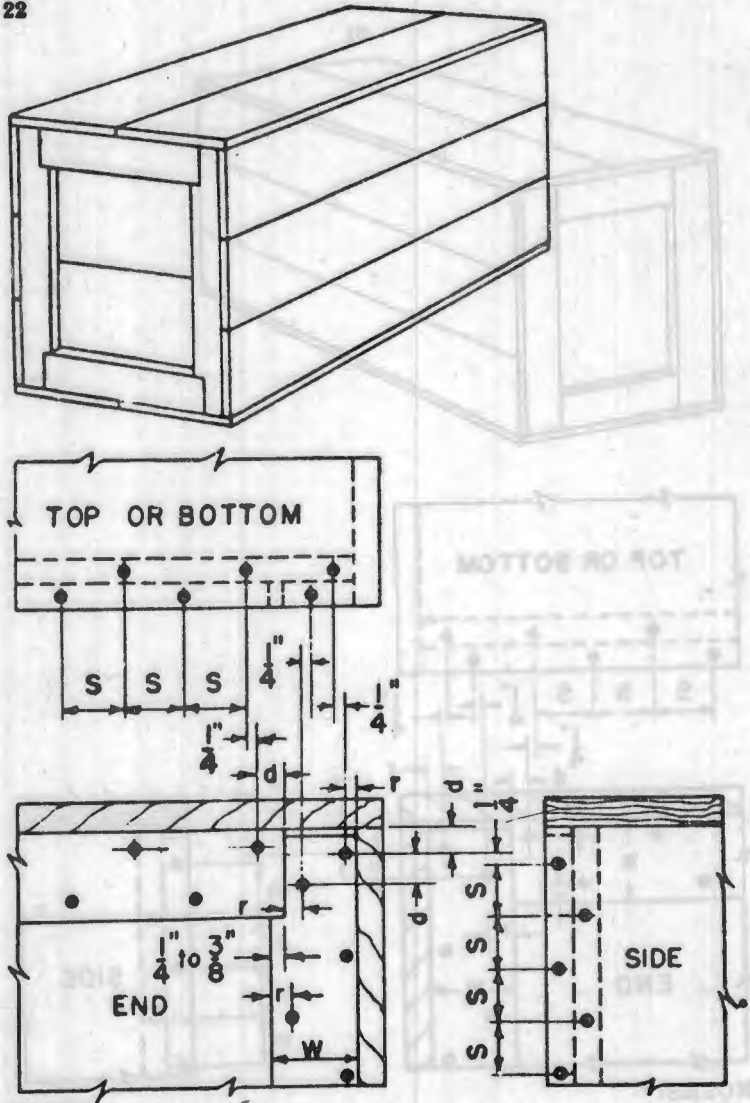


NOTES:
S = Nail spacing. (See par. D–4.)
d = 5/8 inch when ends and cleats are 5/8 inch or less; for thicker lumber d = 3/4 inch.
r = 3/8 inch when W = 2 inches or less; for greater widths, r = 1/2 inch.

FIGURE 2.—*Style 2 box (full cleated ends, butt joints).*

[JAN–J–106A]

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NOTES:

S = Nail spacing. (See par. D–4.)

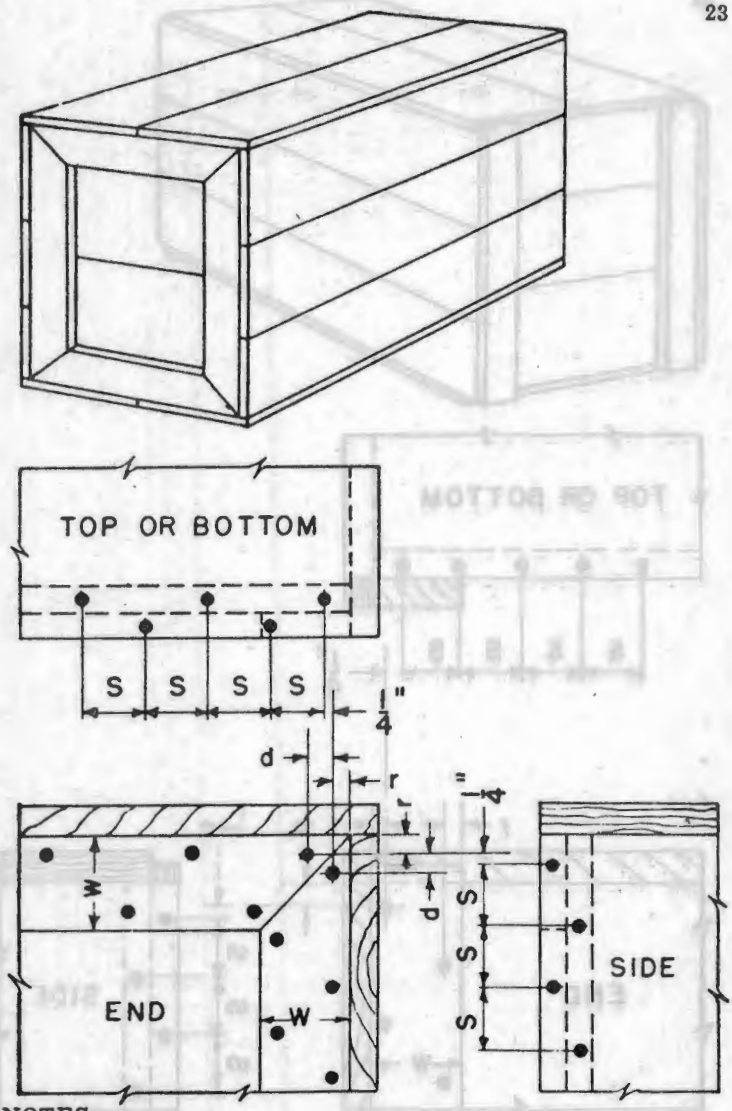
d = 5/8 inch when ends and cleats are 5/8 inch or less; for thicker lumber d = 3/4 inch.

r = 3/8 inch when W = 2 inches or less; for greater widths, r = 1/2 inch.

FIGURE 3.—*Style 2½ box (full cleated ends, notched cleats).*

[JAN–J–106A]

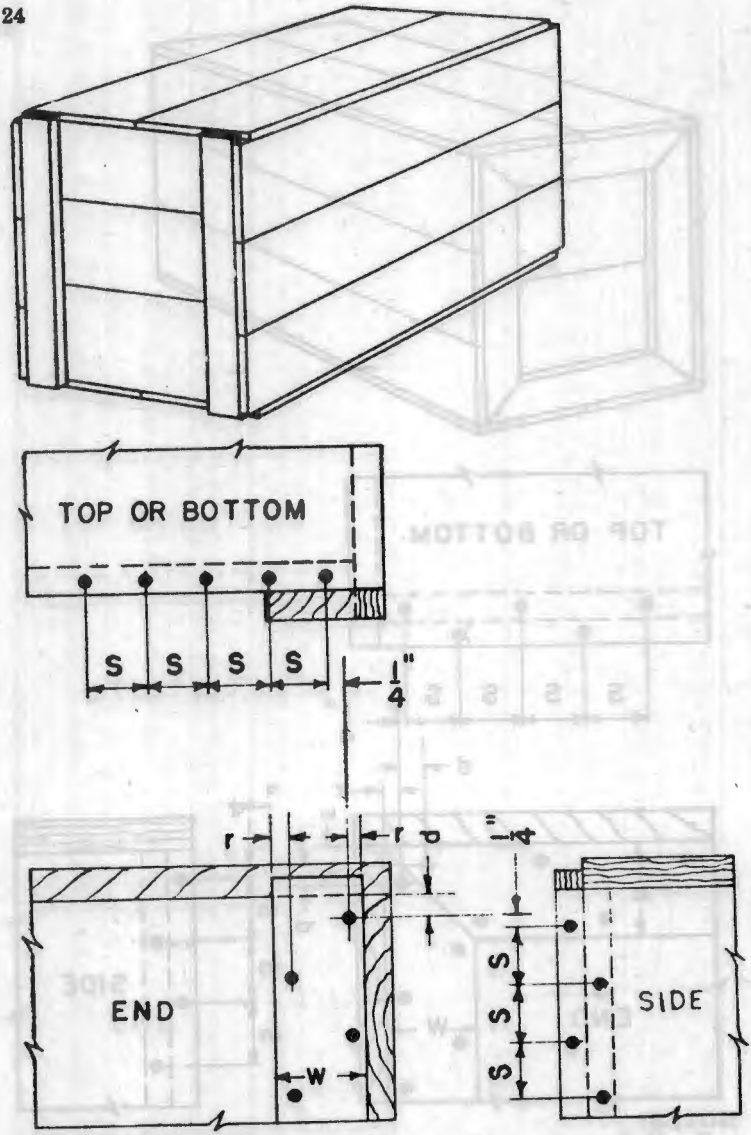
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NOTES:
S = Nail spacing. (See par. D–4.)
d = ⅝ inch when ends and cleats are ⅝ inch or less; for thicker lumber d = ¾ inch.
r = ⅜ inch when W = 2 inches or less; for greater widths, r = ½ inch.

FIGURE 4.—*Style 8 box (full cleated ends, mitred joints).*
[JAN–J–106A]

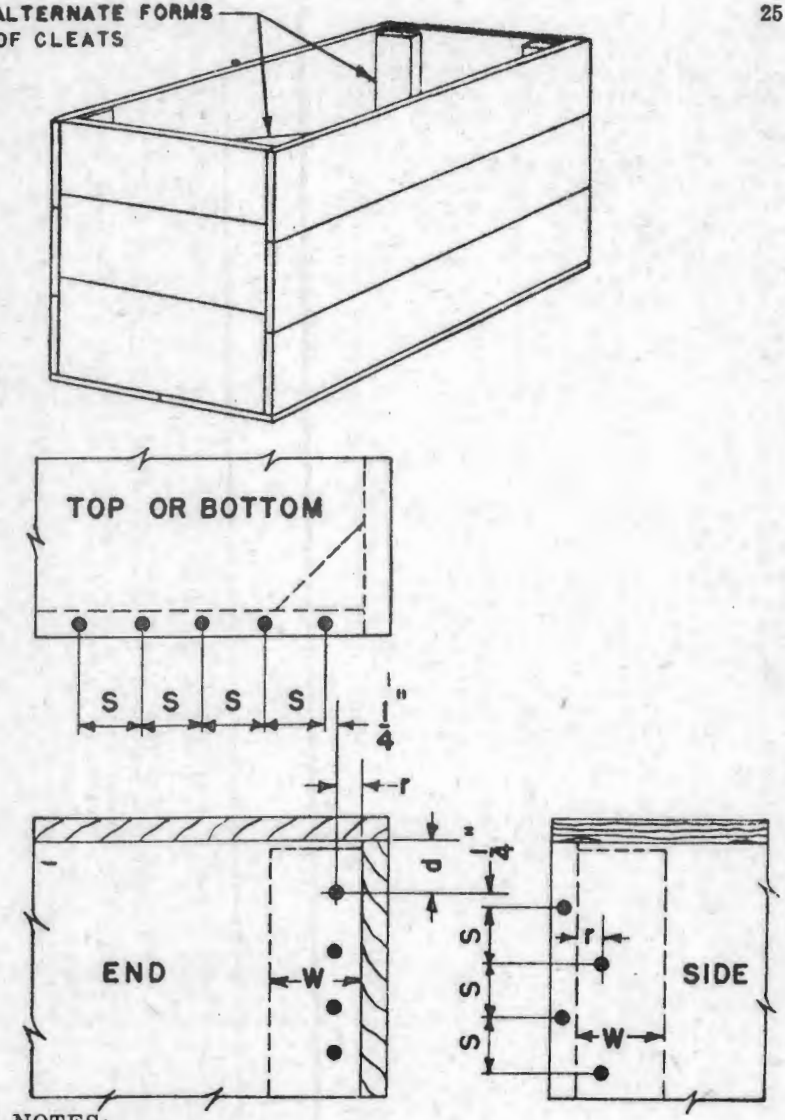
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NOTES:
S = Nail spacing. (See par. D–4.)
d = ⅝ inch when ends and cleats are ⅝ inch or less; for thicker lumber d = ¾ inch.
r = ⅜ inch when W = 2 inches or less; for greater widths, r = ½ inch.

FIGURE 5.—*Style 4 box (two exterior end cleats).* [JAN–J–106A]

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NOTES:

S = Nail spacing. (See par. D–4.)

d = 5/8 inch when ends and cleats are 5/8 inch or less; for thicker lumber d = 3/4 inch.

r = 5/8 inch when W = 2 inches or less; for greater widths, r = 1/2 inch.

When cleats are rectangular, nailing shall be similar to that for style 4 boxes.

FIGURE 6.—*Style 5 box (two interior end cleats).*

[JAN–J–106A]

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